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Lesson Notes

MODULE 1 | LESSON 2

COUNTERPARTIES AND CREDIT RISK

Reading Time	90 minutes
Prior Knowledge	Interest
Keywords	Credit Risk, Collateral, Regulation, Deposit Insurance, Bonds

In the previous lesson, we looked at time value of money and the mathematics of present and forward value together with some elements of interest rate theory. The knowledge of these concepts are essential for analyzing the interaction between a borrower and a lender, but they are also powerful tools that allow us to solve important practical problems. One of the justifications for interest rates usually being higher for longer-duration investments has to do with the risk that the borrower may default.

In this lesson, we examine the process of default and the broader picture of credit risk. We will examine how institutions, regulation, and insurance all help to keep this credit risk in check.

1. Risks in Finance

The world of finance is full of risks. We cannot avoid risks altogether because like we sometimes hear "*no pain, no gain*." In fact, a big part of what a quantitative analyst does deals with recognizing risks, measuring them, and hedging them.

Risk can be divided according to two main categories: **systematic** and **non-systematic**. Systematic risk affects the entire market while non-systematic risk is asset-specific. During your time in the WQU program, you will learn a great deal about risks and how to handle them.

Right now, we will only mention a few main types of risk:

- **Economic risk:** the risk that adverse changes in economic conditions negatively impact businesses, industries, or even entire economies. As investors, we must pay close attention to the economic conditions of those countries in which we plan to invest.
- **Equity risk:** the risk that stock prices will change in an unfavorable manner. If we are investors buying equities, we do so hoping to earn a return that exceeds the risk-free interest rate or else we would not be compensated for taking the risk.

- **Interest rate risk:** the risk that interest rates can change in an unfavorable manner. Imagine a bank that provides long-term mortgages, say 30-year mortgages, and assume that it does so at 5%. Suddenly, interest rates go up to 8%. In this case, the consumer is happy, but the bank not so much. Alternatively, suppose that you have bought a bond that pays an annual coupon of 4.5% and the newly issued bonds pay a coupon of 6.4%. Now, as an investor, you would not be happy, and if you were to sell your bonds, you would have to sell them below par value, which would lead to capital loss.
- **Model risk:** the risk that we use the wrong model to analyze a specific investment. If the model is wrong, it is quite doubtful that our practical solutions are going to be optimal. As quantitative analysts, we must always consider this risk and make sure we employ reliable modeling techniques while being on the lookout for what could go wrong.

The four risks we have listed above are just a few among many more that are possible. In the next sections, we will consider **credit risk** specifically.

2. Defaulting

Before we consider credit risk, we need to understand the concept of defaulting, i.e. not paying in full when a payment is due. In the previous lesson, we realized that not all borrowers are necessarily able to pay their debt. Before a true case of defaulting occurs, it is possible that we run into an **overdue payment** issue, i.e., a borrower fails to make a timely payment of interest, principal, or even both.

It could be a simple case in which the borrower simply pays late. An overdue payment is considered to be any payment that is past a clearly defined and agreeable payment date. Sometimes, late payments occur due to holidays or weekends delaying the processing of payments. There's always the excuse that the payment is lost in the mail. But a payment is overdue because someone simply did not make the payment.

So, what happens when a borrower doesn't pay? Depending on the type of loan, there could be fees that are assessed. Let's look at both sides: from the depositor's perspective and from the perspective of a lender. First, let's consider the depositor's perspective. After the term to maturity, the depositor is expecting to receive both the earned interest and the principal back. For some reason, suppose the bank failed to pay this. Most likely, this would be a bank error and could be fixed quickly. It's unlikely the bank will have to pay more because the fine print that is part of the agreement may not hold them liable for such operational mishaps (operational risk). Nevertheless, banks invest huge amounts of capital to build computational infrastructure to handle the automatic crediting of interest, so these errors are not very common.

Now let's suppose that borrowers are the ones failing to pay on time. Of course, this can be for any number of reasons. In this case, the banks have **grace periods**, which are extra amounts of time to facilitate the payment being processed. If the payment is still not made by the grace period, then there might be extra fees the bank charges, such as late fees, as well as even raising the interest rate for non-payment. Overdue payments are serious and can be the precursor to an actual default. Let us turn to default, the topic of our next section.

Defaulting is much more serious than just being late on a payment. What does it mean to default? Well, borrowing funds has the implied promise that the funds will be returned in full. In addition to the interest payments, the principal repayment is expected. Default occurs when the borrower fails to make some combination of interest payments, principal payments, or both. A default is an overdue payment that remains unpaid. There may even be multiple payments in default. This is particularly important for commercial businesses that may have several debts. Those debts are associated with what are called **financial covenants**. Debt covenants are essentially restrictions that lenders impose on debt holders and that appear in writing in the lending agreements to ensure that a company will operate within certain rules set by the lenders. Lenders do so to avoid situations where a company may be too aggressive in borrowing money and fail to repay them. Often, a default on one loan can trigger a default on all loans with the requests to pay back all of them. When this happens, unless the company has sufficient liquidity, the request could lead to bankruptcy. Hence, defaulting is not to be taken lightly.

We have established that a default occurs when the borrower fails to make payments (interest or principal) to the lender, i.e., the borrower is not owning up to the responsibility that was set out in the terms of the loan. Clearly, this means that the lender is not going to receive their money in full. It is certainly possible that they may receive a partial amount. Can the lender sue the borrower for the balance? Well, virtually every country has laws that protect borrowers who are unable to make full payments by their declaring a type of **bankruptcy**.

Interestingly, the word bankruptcy comes from the Italian term, *banca rotta*, which means "broken bench." At one point in history, these money dealers worked from benches. If a money dealer ran out of money, they would be taken out of business by having their bench broken. With a *banca rotta*, the dealer would no longer be able to conduct business.

So, the bankruptcy/"broken bench" notion is meant to convey that the person is out of business. Fortunately, bankruptcy laws today protect borrowers, which could include small business owners. They may be allowed to continue operating their business, and the amount they are legally required to pay their lenders may be reduced.

While defaulters are generally able to avoid "debtor's prison" (which was also common in history) or serious criminal penalties, they do suffer a financial setback in terms of their credit becoming less favorable.

For example, suppose you are a lender. You are looking to lend money to one of two different people. The first person has an excellent history of making payments. The second person, however, has consistently declared bankruptcy a number of times. Which of the two borrowers do you think would be given a better interest rate? Well, from a bank's perspective, the borrower who defaults poses such uncertainty as to whether the bank receives its funds back. The bank may require a much higher interest rate to offset that greater uncertainty. This is essentially the first real challenge we face as financial practitioners: the challenge of credit risk.

3. Credit Risk

In this section, we introduce credit risk. **Credit risk** is the risk to the lender that the lender does not receive the full amount of principal and interest payments initially spelled out in the loan. Credit risk is a

risk that there will be an expected loss due to the inability or even unwillingness of the borrower to repay their obligations. Credit risk tends to affect the lender and not so much the borrower.

Example. *An investor buys 10,000 shares of company X at a price of \$15 each. Six months later, the price of company X's shares is down to \$12 a share. Furthermore, company X decides not to pay dividends for the foreseeable future. A student reported that this happens because the investor had credit risk since company X fails to pay dividends and did badly considering that its stock price went down. Is the student correct?*

Solution. *No. First of all, dividends are at the discretion of company X and buying shares is nothing like buying company X's bonds where coupon payments are required if promised. Furthermore, company X never promised to guarantee the value of the shareholders' investments in its equity, again unlike bonds, which guarantee that the full principal amount will be repaid.*

Based on the example just discussed, we should note that credit risk requires forms of contractual obligations and should not be confused with market risk or other types of risk.

Now, we will illustrate a new and important concept. Suppose you put \$10,000 in a savings account. You wonder if that money will be there when you wish to withdraw it. You also wonder if the bank will honor the interest payments you are meant to receive. Indeed, there have been times in history where banks have had **bank runs** and were not able to supply the depositors with their own money. Please refer to the required reading for this lesson, which discusses a bank failure nearly half a century ago and how it continues to affect supervision in banks today.

Well, banks do not just keep the money locked in accounts; they need to use it for other purposes. Throughout history, the banks may have invested in opportunities that turned out to be deeply unprofitable and lost much of the depositor's funds. Some depositors who lined up early to withdraw may have received their fair share, but others who came later were left with losses, whether through the loss of value or through some operational mishap.

So how can we minimize this credit risk from the depositor's point of view? Well, many countries today are part of the [International Association of Depositors Insurers, or IADI](#). In the United States, this is handled by the FDIC: Federal Depositary Insurance Corporation. [They maintain a directory of other countries in the program](#). In South Korea, this is handled by the KDIC, the Korea Deposit Insurance Corporation.

These corporations basically guarantee that there are mechanisms, with the backing of the federal government, to ensure that depositors will never lose their deposits (up to a certain amount, at least) even if the bank itself goes bankrupt. This is a very important idea and our first major theme in the course of credit risk. In order for counterparties to engage in transactions, there has to be a clear understanding of the risks of each other's credit. Depositing funds in a bank account is truly one of the safest transactions in the world of finance. The credit risk is the risk that the bank would be unable to pay interest and principal back. These Depositary Insurance corporations reduce the risk by providing backup for those banks. These corporations, with the help of government agencies, will require banks to show prudent use of those funds to avoid situations that might cause them to lose the deposited amount. These corporations ensure that there is widespread trust and confidence in the financial

system. Can you imagine if people put money in banks, then the banks declare bankruptcy and depositors lose their hard-earned savings? People would be afraid to use banks. Banks are the cornerstones of the financial industry.

Let's switch directions and look at credit risk from the bank's point of view when lending to an individual or small business. Unlike banks, an individual borrower does not have the backing of the Insurance Corporation. Instead, a borrower's creditworthiness comes through their reputation to repay their debts in a timely manner. Nevertheless, an individual may have unexpected income loss due to a job loss or simply unforeseen higher expenses like medical expenses. Any circumstance that hinders the ability of the borrower to repay a debt is an example of credit risk.

Credit risk converts much of finance from deterministic problems to random problems. Let's consider the lender's perspective. The amount of total funds received should be known. Several times, we have seen the equation that shows the amount of interest received. Yet, thanks to credit risk, not all payments are guaranteed to be made. They have some uncertainty. There is a small probability associated with the timing and/or the size of each payment. Credit risk is endemic to virtually all transactions where there are counterparties engaged in a transaction. One thing that can help mitigate credit risk is collateral, the topic of our next section.

Example. Suppose that you have \$100,000 deposited with a local bank in the U.S., which is FDIC insured. The local bank has invested heavily in providing residential mortgages. A big economic downturn occurs, and many people lose their jobs and cannot make their mortgage payments. The bank gets in trouble and fails. In doing so, it declares its clients should expect to receive 40 cents on the dollar. How much money should you expect to receive back from your \$100,000 deposited?

SOLUTION. If your account is FDIC insured, then you are protected up to \$250,000; therefore, you should expect to receive back your entire \$100,000. However, if your account were not FDIC insured (not all accounts are treated the same, but savings accounts and checking accounts, the most typical examples, usually are), then you should expect to receive back

$$100,000 \times 40\% = 40,000;$$

i.e., \$40,000.

4. Collateral

Collateral is a financial or physical asset that serves as a backup for the credit risk of the borrower. Suppose a borrower has a \$1,000 loan from a bank. From the bank's point of view, we learned that there's an uncertainty associated with recollecting this \$1,000. How can the bank be expected to bear this risk? One way they could do so is to require collateral for the loan. For example, suppose the loan went to purchase a car. The car serves as collateral. If the borrower continues to make payments in a timely fashion, then the borrower can continue to use the car. However, if the borrower fails to make timely payments, the ownership of the car effectively diverts back to the lender. This transfer is the purpose of collateral. The car collateralizes the loan. Defaulting on the loan means the bank gets the car instead of the money that was lent to buy it. This is an example of collateral that is physical.

Banks don't really want to own cars as they are in the financial industry and not the automotive industry. Banks do want to measure credit risk carefully so their customers can provide timely payments with minimal defaults. This allows them to focus on their core business, such as servicing depository accounts and lending funds to qualified individuals and businesses. Banks are not the only financial institutions that can do this. What other institutions exist in the financial landscape? We will discuss this topic in the next section.

5. Financial Institutions

There are distinct types of financial institutions. One way to consider distinct kinds of financial institutions is to look at the types of activities in which they engage. In doing so, we can categorize financial institutions according to the way in which they provide their primary services:

1. **Depository Institutions.** Depository institutions include banks and their close relatives, such as savings and loan companies, credit unions, trust companies, and mortgage loan companies.
2. **Contractual institutions.** Contractual institutions are those whose businesses provide long-term contractual services. The main types of these are insurance companies and pension funds.
3. **Investment institutions.** Investment institutions are those financial institutions that either originate or manage ongoing investments. These include underwriters, investment banks, mutual funds, closed-end funds, hedge funds, and unit investment trusts.

The reason to know these distinct types of financial institutions is to understand that each offers different services. Keep in mind that some firms can easily be involved in all three services. For example, you may find a multinational bank that can:

- provide you with a loan or mortgage,
- maintain your property insurance program, and
- manage your investment account.

So, within multinational large financial institutions, it is likely that a multitude of services are provided. It is also easy to find companies that specialize in one of the services. For example, within the world of investment institutions, you may find a mutual fund that provides a way to own a collection of bonds that are professionally managed. The mutual fund will not be of any help, however, if you need a loan, a mortgage, or insurance. Furthermore, there are also new FinTech or disruptive types of businesses that are providing these financial services independently. For example, it is possible to find a FinTech company that allows you to transfer money from one place to another ("The 7 Best Transfer Money and Currency Exchange Apps"). Banks charge hefty wiring fees for this service, but FinTech companies, through apps on your smart phone, can do this for a fraction of the cost if not for free. Later in this module, we will distinguish between distinct types of financial institutions to better understand their features and benefits.

Certain large financial institutions are so important to the financial system that their demise could wreak havoc. For example, if a multinational bank were to fail, it could cause widespread fear and panic, leaving depositors without their funds, causing insurance claims to go unpaid, failing pension recipients who need their retirement income, etc. Can we trust these financial institutions to adopt best

practices and ethical behavior? While an amount of self-regulation is helpful, it is even better to have regulation at regional, national, and even international and global levels. This regulation can provide stability and reassurance to promote a rehabilitative reliability and assurance that the financial institution acts in prudent and appropriate ways for the trust of the stakeholders in the system. Prior to the global recession, these institutions ran into trouble and looked for assistance to help bail them out of trouble. This term is known as **"too big to fail."** As a result, indeed, some of them were too big to fail, meaning that their fall could have caused further collapse in the financial system. What helps to minimize or contain this risk? What would help to avoid the problems in the first place? Regulation.

6. Regulation

To understand regulation in the financial industry, first, let us simply think of regulation in a sporting industry. For example, consider football (or soccer as it is known in the United States). There are specific rules in the sport of football. These rules apply to games that are played by most nations in the world. These nations meet periodically on the football field to determine the winner of the World Cup. These rules are well known by players, enforced by referees, and often challenged by the coaches who find that they may have been broken to the disadvantage of their team. Imagine a football game without such rules: the fouls, the cheating, the kicks we don't see, etc. That would be a different game altogether. The rules and regulations are meant to ensure a safe, fair, and equitable game. They also increase reliability by ensuring that the winning team did so without resorting to cheating or using unethical and unscrupulous tactics. That builds confidence in the spectators and sponsors, knowing that the business model of advertising and marketing work in a trustworthy environment.

In the same way, financial institutions have regulation. Financial regulation is meant to ensure the safety and protection of the interests of stakeholders. These stakeholders include consumers, taxpayers, employers and employees, pension holders, investors, homeowners, small businesses, large businesses, and even the financial institutions themselves. This list comprises everyone in the financial system.

Financial regulation not only includes the rules and practices by which firms and individuals operate but also the enforcement of said rules. Just as in the game of football, the rules don't change, but there can be games where the referees selectively enforce rules, miss certain bad plays, or simply make bad calls. Likewise, in finance, some companies break regulatory rules, receive fines or penalties, and fail to comply with mandated policies. Regulation seeks to ensure that the financial system works in a smooth and trustworthy manner.

It is not always easy to determine whose authority a problem may fall under. It may be difficult to identify the culprit in markets with millions, and perhaps billions, of transactions. The complexity of trades, investments, disclosures, and informational asymmetries can create scenarios that have never occurred before. Our goal here is to simply introduce the concept of regulation. In the next topic, we will tackle how regulation can help to minimize credit risk.

7. Insured Accounts

When you think of finance, you can think of it as two-way interactions. Let's examine the “*assurance*” those accounts have from each side. Let us begin with the lender. The lender has lent funds to a borrower. How can the lender have assurance that the borrower will repay? As discussed, one of the contributions of regulation that came out of the global crisis is that there are more transparent qualifications needed to get the loan in the first place. The idea of a no-documentation loan is harder to achieve. Even though this might have been more “profitable” for the bank in the short run, the aggregation of multiple loans is a definite increase in the credit risk of the bank. Therefore, regulation can help assure the stakeholders of that financial institution—its investors, its employees, and even other participants in the financial industry—that the financial institution is acting in a prudent manner in its lending.

Likewise, from the depositor's point of view, how can the depositor be assured that the bank is protecting its deposits? For this, we return to the depository insurance corporation. There are many countries that have a type of deposit insurance system. A list of several of these countries with the max amount insured can be found at the following [link](#) from Wikipedia. These countries provide not merely *assurance* but *insurance* that if anything happens to a deposit (up to a specified monetary amount), then the depository insurance corporation guarantees the amount. Therefore, borrowers can sleep well at night knowing that their deposits are safe. For example, in South Korea, each depositor has protection of up to 20 million Korean won. For each insurance policy holder, there is protection of up to 50 million Korean won. The idea of this insurance coverage helps to provide stability and trust in the financial system. Therefore, the idea of putting your money into a savings account or a certificate of deposit or other insured account is a very safe investment.

But what if you are interested in more than just saving money and earning interest? Is there a way that you could also invest in the fixed-income market? This topic will be addressed in the next lesson: bonds.

8. Basics of Bonds

What is a bond? A bond is a debt security by which the issuer/borrower is committed to paying back to the bondholder/lender the cash amount borrowed (the **principal** or **bond notional**) in addition to periodic interest calculated on the amount of a principal during a given period of time.

Let's see how a bond works with an example.

EXAMPLE *The U.S. Treasury has issued a bond today (1/1/2024) with the following key characteristics:*

- maturity date 12/31/2025;
- coupon rate 4%;
- nominal issued amount \$20,000,000,000; and
- interest payment is semi-annual (typical of U.S. bonds, unlike Euro Treasury bonds that usually pay interest once a year).

Write down the interest payments and the principal repayment with the relative dates.*

SOLUTION

- interest payments:

$$20,000,000,000 \times \frac{4\%}{2} = 400,000,000;$$

i.e., the U.S. Treasury will pay \$400,000,000 every six months, or on 6/30/2024, 12/31/2024, 6/30/2025, and 12/31/2025.

- principal repayment: \$20,000,000,000 on 12/31/2025.

Note that to compute the interest payments for bonds paying semi-annual coupons, we take the coupon rate and divide it by 2.

Why would the U.S. government, via the Treasury Department, or a corporation issue bonds? Well, in the case of the U.S. government, it is a way to finance projects of public interest, like the construction of roads, bridges, and schools. In the case of corporations, it is a way to finance investment projects at a cost (interest rate) that is lower than the return on those investment projects.

We can also look at bonds by saying that a borrower (debtor) has issued an I.O.U. to the bond buyer (creditor). In other words, the borrower has received a stated amount of money up front, the *face value* (also known as notional value) of the bond. In exchange for this principal, the bond issuer commits to providing timely interest payments, known as *coupon* payments, and returning the principal at maturity to the bond buyer.

So, if we wanted to receive some money, we could do one of two things.

1. We could invest our money in a certificate of deposit. In this investment, we are giving money to a bank, which holds it for a specified period of time and then pays us interest and the principal back at the term of maturity.
2. We could purchase a bond. For this, we go to a specific issuer from whom we buy the bond. we might pay them par, or the face value, for a 2-year bond. Then, over the next two years, we would receive coupon payments periodically and the principal back at the end of two years. These amounts are fixed, assuming no defaults.

So, what are the differences between our savings example and the bond example? Both opportunities are examples of **fixed income** investments. It is called fixed income for the two reasons that comprise the term: that the rates being provided are **fixed** and that there is a stream of **income** consisting of the **interest** payments. The repayment of principal is not so much income as it is just the return of your original investment. In other words, so long as the issuer does not default, the bond will repay the amounts just as the bank would repay the amounts.

The difference between the two is that the bond is an asset and not merely an account. The bond can be traded. It is something for which you can see a market price. In the savings example, the CD is not a traded asset. Its value does not fluctuate over its lifetime. You cannot cash it out without severe penalties. The bond, on the other hand, can be sold to someone else, so in a sense, you can get its value liquidated to cash before the term to maturity. That is, the bond does not need to be held to maturity.

To review, there are three ways to get paid from holding a bond:

1. Wait until a coupon date to receive the coupon payment.
2. Wait until maturity at which you receive the principal repayment.
3. Sell the bond before the term to maturity at the prevailing market price.

Since it is an asset, it can be sold in the marketplace. And so, because the bond can trade, its price can fluctuate with the market. What is it that drives the price of bonds? This is the topic of our next lesson.

9. Conclusion

In this lesson, we examined the process of default and the broader picture of credit risk. We looked at how collateral, regulation, and insurance all help to keep this credit risk in check. We ended by introducing bonds. In the next lesson, we'll expand our coverage on bonds.

References

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